Ali Khan

Backend AI Engineer

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PROFILE

Recent Computer Science graduate based in Lucknow with a strong foundation in Python, backend development, and REST API design, complemented by practical exposure to Generative AI and LLM integration through independent projects. Built backend systems leveraging SQL/NoSQL databases, third-party API integrations, and AI-powered workflows including retrieval-augmented generation (RAG) and agentic task automation. Passionate about applying AI/LLM technologies to real backend products, with strong debugging and problem-solving skills and a fast learning curve for new tools and frameworks.

EDUCATION

Sep 2022 – May 2026

Bachelor of Technology, Computer Science and Engineering, Rungta College of Engineering and Technology
Cumulative GPA: 7/10

- Key coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Web Development.
- Built multiple full-stack academic projects, including a Zomato clone using the MERN stack (MongoDB, Express.js, React, Node.js), covering frontend UI, backend REST APIs, and database integration.

Jul 2019 – Jun 2021

Intermediate, TATA DAV Public School
Pursued PCM with Computer Science.

WORK EXPERIENCE

Feb 2025 – May 2025

Data Science Intern, Coding Spoon

- Executed data preprocessing, feature engineering, and model tuning on customer datasets, cutting churn rate by 18%.
- Built and deployed 2+ ML models on 1M+ row datasets, boosting prediction accuracy by 27%.

Aug 2024 – Sep 2024

Data Structure and Algorithm Vocational Training, Coding Spoon

- Completed intensive training in core data structures (arrays, linked lists, trees, graphs) and algorithms (sorting, searching, recursion, dynamic programming).
- Solved 50+ coding problems on LeetCode and HackerRank, improving problem-solving speed and accuracy by about 20%.

PROJECTS

May 2026 – Present

AI-Powered Customer Support Backend

- Developed and deployed a **RESTful backend API using Python (FastAPI)** implementing Retrieval-Augmented Generation (RAG) to answer customer queries by integrating **LLM APIs (OpenAI/Anthropic)** with a document embedding pipeline, reducing manual query resolution time by ~40%.
- Designed a **hybrid SQL (PostgreSQL) and NoSQL (Redis)** data architecture to manage document embeddings, chat sessions, and user authentication, supporting concurrent multi-user chat sessions with sub-second retrieval latency.
- Engineered **confidence-based escalation logic** and **prompt engineering guardrails** to auto-route low-confidence AI responses to human support via **third-party Slack API integration**, improving response reliability and reducing hallucinated answers.

May 2026 – Present

Autonomous AI Task Agent

- Built a **Python-based backend service** enabling natural-language task automation by designing an **LLM-powered planning agent** that decomposes user requests into multi-step actions using tool/function calling (ReAct-style agent architecture).
- Integrated multiple **third-party REST APIs** (GitHub, Slack, SMTP) as callable tools within the AI agent, enabling autonomous execution of real-world workflows (e.g., summarizing issues and posting digests) without manual intervention.
- Implemented **asynchronous task processing** using Celery and Redis with PostgreSQL for persistent job/state tracking, improving system scalability and enabling reliable retry handling for failed agent executions.

CERTIFICATIONS

May 2026

Full Stack Development — Physics Wallah
Completed: May 2026 | Credential ID: 0ff95f48-b8e0-49f5-8173-c6cd064a3272

Verify: [https://tinyurl.com/3xm6235s]

SKILLS

Technical

Python • Shapely • GeoPandas • PostgreSQL • PostGIS • Flask • Node.js • Express • React • Leaflet.js • GeoJSON • Bull • Redis • Chart.js • Recharts • Supabase • Prisma • Docker • Git • Vercel • REST APIs • Generative AI • Large Language Models (LLMs) • SQL • Debugging • Prompt engineering • AI agents • Unit testing • FastAPI • Django • OpenAI API • Anthropic API • Gemini API • LangChain • LangGraph • Pinecone • Qdrant • Weaviate • pgvector

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EXTRACURRICULAR

Feb 2024 – Present

Competitive Programming Lead, Google Developers Group
Directed Data Structures and Algorithms training sessions, mentoring over 50 peers to improve problem-solving skills by 30% through targeted exercises.

Mar 2023 – Present

Coordinator, Infinity Coders Club
Orchestrated 50+ weekly programming events and hosted weekly events engaging 100+ participants in computer science discussions and activities, boosting participation rates by 35%.

Dec 2022 – Present

Community Manager, Open-Source Community
Managed a community for programming guides, increasing contributions by 45% and improving project outreach across platforms.